2.37 Integration: Lengths

Worked Solutions

1a)

For $y = \frac{2}{3}x^{3/2}$, $dy/dx = \sqrt{x}$. Thus

$$s = \int_8^{35} \sqrt{1+x}\,dx = \left[\frac{2}{3}(1+x)^{3/2}\right]_8^{35} = 144 - 18 = 126 \text{ units.}$$

1b)

For $y = \frac{4}{3}x^{3/2}$, $dy/dx = 2\sqrt{x}$. Thus

$$s = \int_1^4 \sqrt{1+4x}\,dx = \left[\frac{1}{6}(1+4x)^{3/2}\right]_1^4 = \frac{17\sqrt{17} - 5\sqrt{5}}{6} \text{ units.}$$

2)

For $y = \cosh x$, $dy/dx = \sinh x$, so the speed is $\sqrt{1+\sinh^2 x} = \cosh x$. Using the x-coordinates stated in the question,

$$s = \int_2^3 \cosh x\,dx = [\sinh x]_2^3 = \sinh 3 - \sinh 2.$$

3a)

$$\begin{aligned}\int \sec x\,dx &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\,dx \\ &= \ln|\sec x + \tan x| + c,\end{aligned}$$

since the numerator is the derivative of the denominator.

3b)

For $y = \ln(\cos x)$, $dy/dx = -\tan x$, and the speed is $\sqrt{1+\tan^2 x} = \sec x$ on the given interval. Hence

$$\begin{aligned}s &= \int_{\pi/4}^{\pi/3} \sec x\,dx \\ &= [\ln|\sec x + \tan x|]_{\pi/4}^{\pi/3} = \ln\left(\frac{2+\sqrt{3}}{1+\sqrt{2}}\right) \text{ units.}\end{aligned}$$

4)

Here $dy/dx = x$, so

$$s = \int_0^2 \sqrt{1+x^2}\,dx.$$

Let $x = \sinh u$, so $dx = \cosh u\,du$. The limits are $u = 0$ and $u = \operatorname{arsinh} 2 = \ln(2+\sqrt{5})$. Therefore

$$s = \int_0^{\ln(2+\sqrt{5})} \cosh^2 u\,du$$

$$= \left[\frac{u}{2} + \frac{1}{4}\sinh 2u\right]_0^{\ln(2+\sqrt{5})}$$
$$= \sqrt{5} + \frac{1}{2}\ln(2+\sqrt{5}) \text{ units.}$$

5)

Since $dx/dy = \sinh(y/2)$, the arc length is

$$s = \int_0^{\ln 4} \sqrt{1+\sinh^2(y/2)}\, dy = [2\sinh(y/2)]_0^{\ln 4} = \frac{3}{2} \text{ units.}$$

6a)

The cable passes through $B = (0, 15)$, so $15 = k\cosh 0 - p = k - p$.

6b)

From part (a), $p = k - 15$. At $(70, 40)$,

$$40 = k\cosh\left(\frac{70}{k}\right) - (k - 15),$$

so $f(k) = k\cosh(70/k) - k - 25 = 0$. Evaluating near the root gives $f(101.905) > 0$ and $f(101.915) < 0$. Since f is continuous, $k = 101.91$ to two decimal places.

6c)

Using $k = 101.91$ and $p = 86.91$, the gradient is $dy/dx = \sinh(x/101.91)$. Therefore

$$s = \int_{-70}^{70} \sqrt{1+\sinh^2(x/101.91)}\, dx$$
$$= [101.91\sinh(x/101.91)]_{-70}^{70}$$
$$\approx 151.271 = 151 \text{ m to the nearest metre.}$$

7a)

The minimum occurs at $x = 250$ and has height 60, so $p + 1 = 60$ and $p = 59$. At the minimum, $250/q - r = 0$. At $x = 0$, $y = 80$, so $\cosh r = 21$. Hence

$$r = \operatorname{arcosh} 21 = \ln(21 + 2\sqrt{110}) \approx 3.74, \qquad q = \frac{250}{r} \approx 66.9.$$

7b)

Using the unrounded values $q \approx 66.897$ and $r \approx 3.7371$, the arc length is

$$s = \int_0^{500} \sqrt{1 + \left(\frac{1}{q}\sinh\left(\frac{x}{q} - r\right)\right)^2}\, dx \approx 503.225 = 503 \text{ m to the nearest metre.}$$

8a)

The curve reaches its minimum at $x = 1.6$, $y = 0$. Thus $p - 2 = 0$, so $p = 2$ and $1.6q - r = 0$. At $x = 0$, $y = 1.5$, giving $\cosh r = 1.75$. Therefore

$$r = \operatorname{arcosh}(1.75) \approx 1.15881, \qquad q = \frac{r}{1.6} \approx 0.7243.$$

8b)

Substituting $x = 3$ gives $y = 2\cosh(0.724256 \cdot 3 - 1.15881) - 2 \approx 1.11927$. Hence $CD = 1.12$ m to the nearest centimetre.

8c)

Using the rounded parameters in the model, numerical integration gives

$$\begin{aligned} s &= \int_0^3 \sqrt{1 + (1.44851 \sinh(0.724256x - 1.15881))^2}\, dx \\ &\approx 4.1795 = 4.18 \text{ m to the nearest centimetre.} \end{aligned}$$

9a)

For $y = \frac{1}{k}\cosh^2(kx)$, $dy/dx = \sinh(2kx)$. Therefore

$$s = \int_0^1 \sqrt{1 + \sinh^2(2kx)}\, dx = \left[\frac{1}{2k}\sinh(2kx)\right]_0^1 = \frac{\sinh(2k)}{2k}.$$

9b)

For $y = \frac{1}{k}\sinh^2(kx)$, $dy/dx = \sinh(2kx)$ as well. Thus

$$s = \int_0^1 \sqrt{1 + \sinh^2(2kx)}\, dx = \frac{\sinh(2k)}{2k}.$$

10a)

Let $x = \frac{1}{6}\sinh u$. Then $dx = \frac{1}{6}\cosh u\, du$ and $\sqrt{1 + 36x^2} = \cosh u$. Hence

$$\begin{aligned} \int \sqrt{1 + 36x^2}\, dx &= \frac{1}{6}\int \cosh^2 u\, du \\ &= \frac{1}{12}u + \frac{1}{24}\sinh 2u + c \\ &= \frac{1}{12}\operatorname{arsinh}(6x) + \frac{1}{2}x\sqrt{1 + 36x^2} + c. \end{aligned}$$

10b)

For $y = 3x^2$, $dy/dx = 6x$. Thus

$$\begin{aligned} s &= \int_0^{1/3} \sqrt{1 + 36x^2}\, dx \\ &= \left[\frac{1}{12}\operatorname{arsinh}(6x) + \frac{1}{2}x\sqrt{1 + 36x^2}\right]_0^{1/3} \\ &= \frac{1}{12}\ln(2 + \sqrt{5}) + \frac{\sqrt{5}}{6} \text{ units.} \end{aligned}$$

11a)

Let $x = \sinh^2 u$. Then $dx = 2\sinh u \cosh u \, du$, giving

$$\begin{aligned}\int \sqrt{\frac{x+1}{x}}\, dx &= \int 2\cosh^2 u \, du = u + \frac{1}{2}\sinh 2u + c \\ &= \operatorname{arsinh}\sqrt{x} + \sqrt{x(x+1)} + c.\end{aligned}$$

11b)

For $y = \frac{2}{3}\sqrt{x}$, $dy/dx = 1/(3\sqrt{x})$. With $u = 9x$,

$$\begin{aligned}s &= \int_1^4 \sqrt{1 + \frac{1}{9x}}\, dx = \frac{1}{9}\int_9^{36} \sqrt{\frac{u+1}{u}}\, du \\ &= \frac{1}{9}\left[\operatorname{arsinh}\sqrt{u} + \sqrt{u(u+1)}\right]_9^{36} \\ &= \frac{1}{9}\ln\left(\frac{6+\sqrt{37}}{3+\sqrt{10}}\right) + \frac{2\sqrt{37}}{3} - \frac{\sqrt{10}}{3} \text{ units.}\end{aligned}$$

12a)

For $y = 5 - \ln(1 - x^2)$,

$$\frac{dy}{dx} = \frac{2x}{1-x^2}.$$

Then

$$1 + \left(\frac{dy}{dx}\right)^2 = \frac{(1-x^2)^2 + 4x^2}{(1-x^2)^2} = \frac{(1+x^2)^2}{(1-x^2)^2}.$$

For the stated interval, the arc-length integrand is therefore $(1 + x^2)/(1 - x^2)$.

12b)

Using partial fractions,

$$\frac{1+x^2}{1-x^2} = -1 + \frac{1}{1-x} + \frac{1}{1+x}.$$

Thus

$$s = \left[-x - \ln(1-x) + \ln(1+x)\right]_0^{1/2} = \ln 3 - \frac{1}{2} \text{ units.}$$

13)

For $y = \frac{1}{3}x^{3/2}$, $dy/dx = \frac{1}{2}\sqrt{x}$. Hence

$$\begin{aligned}s &= \int_0^k \sqrt{1 + \frac{x}{4}}\, dx = \left[\frac{1}{3}(4+x)^{3/2}\right]_0^k \\ &= \frac{1}{3}(4+k)^{3/2} - \frac{8}{3}.\end{aligned}$$

Setting $s = 39$ gives $(4 + k)^{3/2} = 125$, so $4 + k = 25$ and $k = 21$ m.

14a)

The minimum at $x = 30$ gives $30/b - c = 0$ and $12 = a + 1$, so $a = 11$. At $x = 0$, $20 = 11 + \cosh c$, hence

$$c = \operatorname{arcosh} 9 = \ln(9 + \sqrt{80}), \qquad b = \frac{30}{\ln(9 + \sqrt{80})}.$$

14b)

Since $dh/dx = b^{-1} \sinh(x/b - c)$, numerical integration gives

$$s = \int_0^{60} \sqrt{1 + \frac{1}{b^2} \sinh^2\left(\frac{x}{b} - c\right)}\, dx \approx 63.448 = 63 \text{ m to the nearest metre.}$$

15a)

$$\frac{dx}{dt} = 2t, \qquad \frac{dy}{dt} = t^2 - 1.$$

Therefore

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 4t^2 + (t^2 - 1)^2 = (t^2 + 1)^2.$$

15b)

$$s = \int_0^4 \sqrt{(t^2 + 1)^2}\, dt = \int_0^4 (t^2 + 1)\, dt = \left[\frac{1}{3}t^3 + t\right]_0^4 = \frac{76}{3} \text{ units.}$$

16a)

$$\frac{dx}{dt} = 2t^2, \qquad \frac{dy}{dt} = 4\sqrt{2}\, t.$$

Thus

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 4t^4 + 32t^2 = 4t^2(t^2 + 8),$$

$$s = \int_0^2 \sqrt{4t^2(t^2 + 8)}\, dt = \int_0^2 2t\sqrt{t^2 + 8}\, dt.$$

Hence $k = 8$.

16b)

$$s = \int_0^2 2t\sqrt{t^2 + 8}\, dt = \left[\frac{2}{3}(t^2 + 8)^{3/2}\right]_0^2$$

$$= 16\sqrt{3} - \frac{32\sqrt{2}}{3} \text{ units.}$$

17)

$$\frac{dx}{dt} = \frac{5}{2}(1+t)^{-1/2}, \qquad \frac{dy}{dt} = -\frac{5}{2}(1-t)^{-1/2}.$$

Therefore

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \frac{25}{4(1+t)} + \frac{25}{4(1-t)} = \frac{25}{2(1-t^2)}.$$

The arc length is

$$s = \int_0^1 \frac{5\sqrt{2}}{2\sqrt{1-t^2}}\,dt = \frac{5\sqrt{2}}{2}[\arcsin t]_0^1 = \frac{5\sqrt{2}\pi}{4} \text{ units.}$$

18)

Since $x = r\cos\theta$ and $y = r\sin\theta$,

$$\frac{dx}{d\theta} = -r\sin\theta, \qquad \frac{dy}{d\theta} = r\cos\theta.$$

Thus the speed is

$$\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = \sqrt{r^2(\sin^2\theta + \cos^2\theta)} = r.$$

Hence

$$s = \int_0^{2\pi} r\,d\theta = 2\pi r.$$

19)

Differentiating the parametric equations,

$$\frac{dx}{dt} = -2e^{-t}\sin t, \qquad \frac{dy}{dt} = -2e^{-t}\cos t.$$

Therefore the speed is $2e^{-t}$ and

$$s = \int_0^3 2e^{-t}\,dt = [-2e^{-t}]_0^3 = 2 - \frac{2}{e^3} \text{ units.}$$

20a)

$$\frac{dx}{dt} = -3\sin t, \qquad \frac{dy}{dt} = 3 + 3\cos t.$$

Therefore

$$\begin{aligned}\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= 9\sin^2 t + 9(1+\cos t)^2 \\ &= 18(1+\cos t) = 36\cos^2\left(\frac{t}{2}\right).\end{aligned}$$

20b)

For $0 \le t \le \pi$, $\cos(t/2) \ge 0$. Thus

$$s = \int_0^\pi 6\cos\left(\frac{t}{2}\right)dt = \left[12\sin\left(\frac{t}{2}\right)\right]_0^\pi = 12 \text{ units.}$$

21)

$$\frac{dx}{dt} = 9\sin^2 t\cos t, \qquad \frac{dy}{dt} = -9\cos^2 t\sin t.$$

Then

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 81\sin^2 t\cos^2 t(\sin^2 t + \cos^2 t)$$
$$= \frac{81}{4}\sin^2(2t).$$

For $0 \le t \le \pi/2$, the speed is $\frac{9}{2}\sin 2t$. Therefore

$$s = \int_0^{\pi/2} \frac{9}{2}\sin 2t\, dt = \frac{9}{2} \text{ units.}$$

22a)

$$\frac{dx}{dt} = 4\cos t - 4\cos 4t, \qquad \frac{dy}{dt} = -4\sin t + 4\sin 4t.$$

Using $\cos A\cos B + \sin A\sin B = \cos(A-B)$,

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 16 + 16 - 32\cos(4t - t)$$
$$= 32(1 - \cos 3t) = 64\sin^2\left(\frac{3t}{2}\right).$$

22b)

For $0 \le t \le \pi/3$, $\sin(3t/2) \ge 0$. Hence

$$s = \int_0^{\pi/3} 8\sin\left(\frac{3t}{2}\right) dt = \left[-\frac{16}{3}\cos\left(\frac{3t}{2}\right)\right]_0^{\pi/3} = \frac{16}{3} \text{ units.}$$

23)

$$\frac{dx}{dt} = k(1 + \cos t), \qquad \frac{dy}{dt} = k\sin t.$$

Therefore

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = k^2(2 + 2\cos t) = 4k^2\cos^2\left(\frac{t}{2}\right).$$

For $0 \le t \le \pi$, the speed is $2k\cos(t/2)$, and

$$s = \int_0^{\pi} 2k\cos\left(\frac{t}{2}\right) dt = 4k.$$

Since $s = 12$, $k = 3$.

24a)

Let $u = \sinh x$. Then $du = \cosh x\, dx$ and $\cosh^2 x = 1 + u^2$. Therefore

$$\int \operatorname{sech} x\, dx = \int \frac{du}{1+u^2} = \arctan u + c = \arctan(\sinh x) + c.$$

24b)

$$\frac{dx}{dt} = -\tanh t \operatorname{sech} t, \qquad \frac{dy}{dt} = \operatorname{sech}^2 t.$$

Using $\tanh^2 t + \operatorname{sech}^2 t = 1$, the speed is $\operatorname{sech} t$. Thus

$$s = \int_0^{\ln 2} \operatorname{sech} t \, dt = [\arctan(\sinh t)]_0^{\ln 2} = \arctan \frac{3}{4} \text{ units.}$$

25a)

Let $x = \sinh u$, so $dx = \cosh u \, du$ and $\sqrt{1+x^2} = \cosh u$. Then

$$\begin{aligned} \int \sqrt{1+x^2} \, dx &= \int \cosh^2 u \, du = \frac{1}{4} \sinh 2u + \frac{1}{2} u + c \\ &= \frac{1}{2} x \sqrt{x^2+1} + \frac{1}{2} \operatorname{arsinh} x + c \\ &= \frac{1}{2} x \sqrt{x^2+1} + \frac{1}{2} \ln\left(x + \sqrt{x^2+1}\right) + c. \end{aligned}$$

25b)

$$\frac{dx}{dt} = 2at, \qquad \frac{dy}{dt} = 4a,$$

so the speed is $2a\sqrt{t^2+4}$. Therefore

$$\begin{aligned} s &= \int_0^1 2a\sqrt{t^2+4} \, dt \\ &= a\left[t\sqrt{t^2+4} + 4\ln\left(t + \sqrt{t^2+4}\right)\right]_0^1 \\ &= a\left(\sqrt{5} + 4\ln\frac{1+\sqrt{5}}{2}\right) \text{ units.} \end{aligned}$$